

Project Idea

Your group must agree on one simple web-based system.

Examples (pick one or something similar):

- Password manager
- Favorite links storage system
- Medical form reuse system

Rules for the project idea:

It must:

- Require user authentication (login system)
- Use a web form connected to a database (store + retrieve data)
- Include security best practices (hashing, encryption)
- Have help documentation (PDF mockup is fine)

Deliverable:

A short paragraph describing:

- What the system is | Clicking game |
- Who it is for | Anyone who is bored |
- What problem it solves | Boredom |

Define Project Scope

Before writing requirements, clearly define:

- What the system will do
- What it will NOT do

Example:

- Will support basic user login and data storage
- Will NOT include mobile apps or third-party integrations

Deliverable:

A short "Project Scope" section in the document.

Requirements Phase

3A. Create Personas

You must create at least two personas:

1. User Persona

- Who they are
- What they want to accomplish
- Their technical comfort level
- Their frustrations/pain points

2. Hacker Persona

- What kind of attacker they are
- What they want to exploit
- How they might try to abuse the system

Deliverable:

A subsection titled:

- “User Persona”
- “Hacker Persona”

3B. Create User Stories

User stories explain what the system should do, written from the user’s perspective.

Format:

As a <user>, I want <feature> so that <benefit>.

Examples:

- As a user, I want to log in securely so my data is protected.
- As a user, I want to save and retrieve my data easily.

Deliverable:

A bulleted or numbered list of user stories.

3C. Functional Requirements

Translate user stories into clear system requirements.

Examples:

- The system shall require users to authenticate before accessing data.
- The system shall store user data in a database.
- The system shall allow users to retrieve stored data.

Deliverable:

A “Functional Requirements” section with numbered requirements.

3D. Security Requirements (Required)

You must explicitly address security.

Include:

- Password hashing (not plain text)
- Encryption for sensitive data
- Protection against common attacks (e.g., brute force, SQL injection)
- Any regulatory considerations (HIPAA, FERPA, etc., if applicable)

You may note limitations if full compliance is not realistic.

Deliverable:

A “Security Requirements” section.

3E. Documentation & Mockups

You must show how the system would look and be used.

This can include:

- Login screen mockup
- Form screen mockup
- Help documentation outline
- PDF showing example screens

You do NOT need real screenshots—mockups are acceptable.

Deliverable:

- Images or diagrams
- Short explanations of each screen

Design Phase

Design Phase

This is about structure, not full code.

4A. Identify System Components

Decide whether your design will be:

- Class-based (OOP)
- Function-based

Then define:

- Class names or function names
- Attributes/parameters
- Return values
- Data types

Important:

Only create headers, not full implementations.

Deliverable:

A section listing:

- Classes or functions
- Their attributes and methods
- Return types

4B. Call Hierarchy (Recommended)

Show how components interact:

- Which function calls which
- Order of operations

This can be a diagram or bullet flow.

Deliverable:

A simple call hierarchy diagram or explanation.

4C. UML Diagrams (REQUIRED)

At minimum:

- Class Diagram (most common)

Optional:

- Use Case Diagram
- Sequence Diagram

Deliverable:

UML diagram(s) embedded in the document or attached.

Personas

User Persona:

Persona Name

Johnny Adams

Demographics

- Age: 19
- Occupation / Role: Student
- Education level: College
- Location / Environment: Home PC/Mobile Phone

Background

Johnny is a casual gamer that enjoys low-stress games. They enjoy incremental clicking games due to the lack of intense focus needed to progress.

Goals

- Accumulate points
- Beat highscore
- Compete on the leaderboard

Motivations

- Sense of achievement
- Quick dopamine hits
- Competition

Behaviors

- Clicks actively during free time
- Actively trains hand-eye coordination
- Aims to beat personal highscore
- Competitively progresses on leaderboard

Needs

- Simple interface
- Gradual progression
- Increasing intensity
- Multiplayer leaderboard

Pain Points / Frustrations

- Decreasing timer
- Not beating highscore
- Poor hand eye coordination

Technical Skill Level

Intermediate

Usage Context / Scenario

Johnny logs in while waiting for class to start, plays a few rounds, purchases an upgrade, and checks their leaderboard position before logging off.

Hacker (Threat Actor) Persona:

Persona Name

Clicklord

Threat Actor Type

Script Kiddie

Skill Level

Low to Medium

Background

ClickLord is a player who enjoys finding shortcuts and gaining an unfair advantage in online games. They are not a professional hacker but are familiar with browser development tools, scripts and basic automation. Their focus is gaining an unfair advantage rather than causing system damage.

Primary Goals

- Artificially inflate click counts
- Reach the top of the leaderboard
- Bypass game limitations

Motivations

- Leaderboard recognition
- Desire to “game the system”
- Curiosity as to how the game functions

Common Techniques / Behaviors

- Automated clickers/autoclickers
- Manipulating client-side values

Target Selection Criteria

- Lack of anti-cheat mechanisms
- Games with client-side scoring

Constraints / Risks

- Risk of detection/ban
- Limited technical knowledge

Typical Attack Scenario